

A tree-community-level analysis of successional status and gap-phase and postfire regeneration of range-margin *Thuja plicata* (western redcedar)

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Abstract: Range-margin populations may have different life history characteristics than other populations, and the montane forests of Glacier National Park represent the easternmost limit of several western conifer species including *Thuja plicata* Donn ex D. Don (western redcedar) and *Tsuga heterophylla* (Raf.) Sarg. (western hemlock). Understanding regeneration strategies of range-margin populations is important for predicting species distribution shifts. We identify successional status, seedling substrate preferences, and the degree to which different species establish within canopy gaps and the forest matrix. *Thuja plicata* and *Tsuga heterophylla* were each found to comprise at least 35% of all tree size classes, with six other species each contributing less than 6%. Similarly, other species each comprised less than 5% of canopy gap and understory seedlings. *Thuja plicata* is typically thought of as a late-successional species but it, along with *Tsuga heterophylla*, dominated stands during all stages of successional development. Similarly, they dominated both closed forest and gap establishment sites on all rooting substrates but preferred wood for establishment. Even with the likelihood of increased fire frequency, these range-margin populations likely have the capacity to maintain their current distributions. As disturbance regimes are modified across landscapes, it may be possible to observe the potential to adapt to local conditions in other range-margin populations.

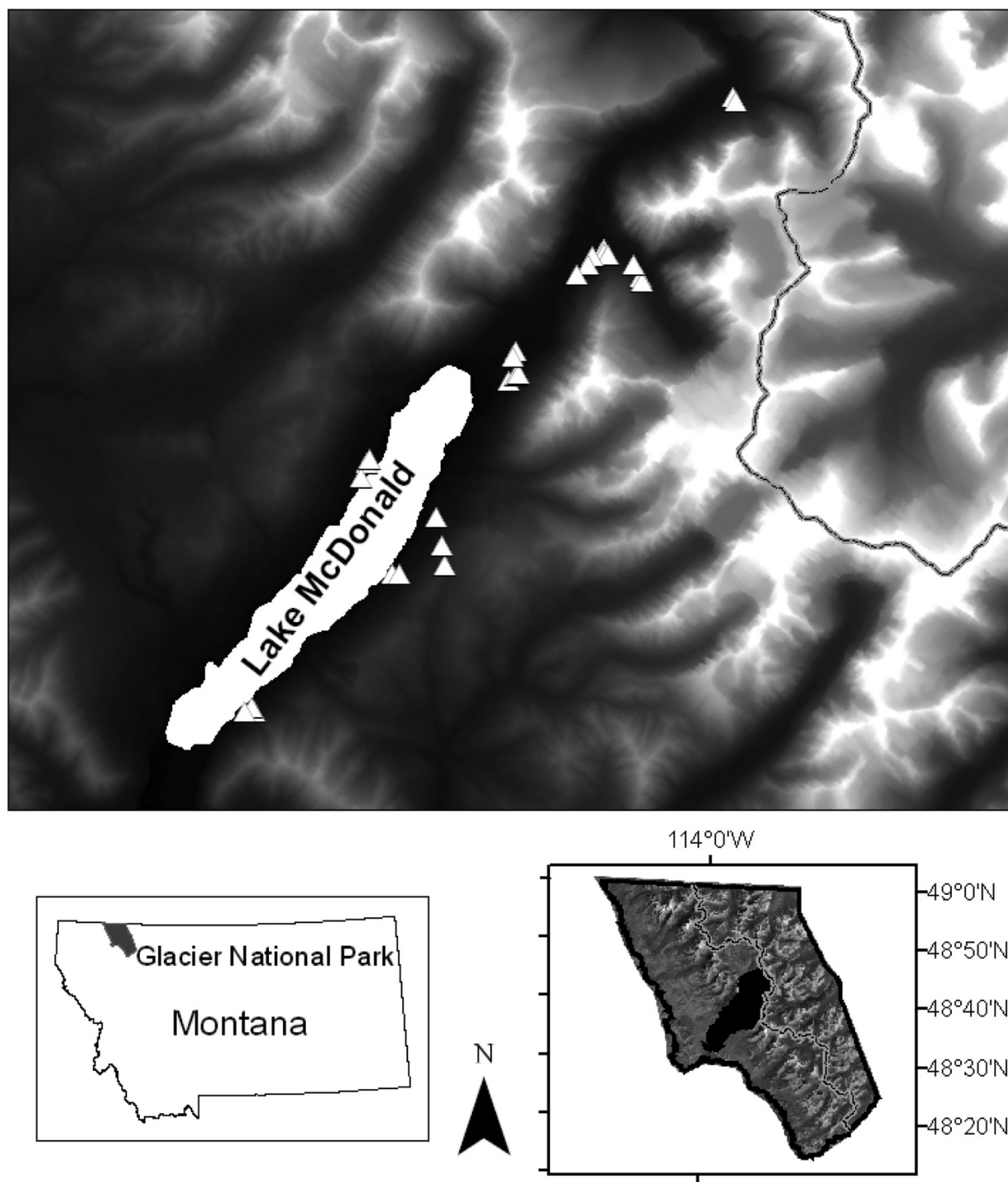
Résumé : Les populations en marge de leur aire de répartition peuvent avoir un cycle biologique dont les caractéristiques diffèrent de celles des autres populations et les forêts montagnardes du parc national des Glaciers correspondent à la limite orientale de plusieurs espèces de conifère de l'ouest, dont *Thuja plicata* Donn ex D. Don (thuya géant) et *Tsuga heterophylla* (Raf.) Sarg. (pruche

concentrations of available nutrients, decayed coarse wood on the forest floor can act as an establishment site for tree seedlings (Harmon and Franklin 1989; Takahashi et al. 2000). Its distribution and nutrient content can vary along topographic and temporal gradients (Laiho and Prescott 2004). Many tree species such as *Thuja plicata* and *Tsuga heterophylla* (Raf.) Sarg. (western hemlock) frequently establish in higher numbers on large decaying logs than on the surrounding forest floor (Harmon 1987; Narukawa et al. 2003).

Both microenvironment and substrate quality change as a result of disturbance within forests (Barik et al. 1992; Zhu et al. 2007), which can range from large-scale, stand-replacing fires to the falling of a single tree, creating a gap in the forest canopy. Gap-phase regeneration is one process by which forest structure is maintained and in mature forests may be the primary form of tree recruitment to the canopy, either from new seedlings or from seedlings already established beneath the canopy (Gray and Spies 1996; Kimmins 2004). This is particularly true for shade-tolerant species, which may persist in a suppressed state for decades beneath a closed canopy (D.A. McKenzie, personal observation). Creation of canopy gaps may also provide the opportunity for establishment of either the dominant tree species or different species that are scarce within a stand, potentially altering stand composition over time (Lertzman 1992). Forest canopy gaps also provide different microsite conditions including available light, moisture, or soil temperature (Zhu et al. 2007).

Thuja plicata is commonly found in two somewhat disjunct locations in North America; coastal populations in the Pacific Northwest (PNW) and inland populations along the Northern Rocky Mountains (US Geological Survey 1999). It reaches its easternmost distribution in GNP, Montana (Tesky 1992). In more western populations, it frequently establishes on decaying logs and stumps (Christy and Mack 1984), and canopy gaps can represent the primary location of tree establishment by sexual reproduction in mature forests (Wright et al. 1998; Larson and Franklin 2005). In moist, mature PNW forest stands, decaying coarse wood can be found in large amounts as a result of tree mortality

Fig. 1. Map of the study location. The lower left map shows the location of Glacier National Park (shaded area) within the state of Montana, USA. The lower right map shows the location (shaded area) of the McDonald drainage within Glacier National Park in which the study took place. Finally, the upper map shows the location of our study transects within the McDonald drainage. Triangles represent transect start-point locations; the broken line represents the Continental Divide. Dark areas represent lower elevations and light areas represent high elevations.



height but not extending into the canopy; Harmon and Franklin 1989) species, size, and density and seedling (<1 cm diameter at breast height) species and density were recorded on belts along the transects (4 m wide for trees/subcanopy trees, and 2 m wide for seedlings). The growing substrate of seedlings was recorded, including the decay class, if the substrate was wood (Narukawa et al.

2003). In addition to the size (diameter at breast height) of the trees, we made assumptions about the life stage of the trees in relation to stand age. Lacking definitive age structure data because of heart rot found in many trees, we placed trees into height classes identified as fire survivor (living legacy), seedling, subcanopy, and canopy trees. We based our classification on tree size

(bole diameter), canopy position (height and canopy depth), and degree of self-pruning. Trees identified as part of the initial post-fire tree cohort (canopy class) were those that showed evidence of self-pruning that results from stem exclusion as a single-aged, postfire stand matures (Klinka and Brisco 2009). To identify the proportion of the stand to which our gap results were relevant, we also calculated the area of the forest canopy that was represented by gaps along the 150 m transects (Runkle 1992).

Gaps chosen for sampling varied in size but were relatively small, having resulted from the death of one to three canopy trees. They were young enough to not yet have been completely filled by ingrowth, as evidenced by a lack of canopy above the stump of the gap maker, and were still identifiable as a gap. We did not identify the cause of death for the trees that created the gaps, but these causes were likely varied and could include pathogens, senescence, or random events such as lightning strikes. Edaphic and topographic gaps (water, rock gaps) were excluded from gap-level vegetation sampling but were included in the calculation of total canopy gap area for the stand. Only the first gap that met the above criteria was sampled per 150 m transect. Within each sampled gap, 2 m wide belt transects were used to quantify the gap fillers (tree seedlings and subcanopy saplings with the potential to attain the canopy), and 0.25 m² quadrats placed each metre along the belts were used to characterize substrates in which seedlings established and to quantify understory cover categories. Transect belts and quadrats within each gap were arranged to include the longest segments along the cardinal directions and stretched the length of the expanded gap edge (circumference described by the base of surrounding matrix trees). The length of these transects varied and was used to calculate area, which, along with surrounding canopy height, affects gap light, temperature, and ambient moisture conditions affecting the forest floor (Lawton and Putz 1988). To identify substrate preference of seedlings, rooting substrate was recorded in both matrix and gap belts for each seedling encountered. Seedling rooting substrate categories used were log, soil (including moss and lichen), and stumps. Taxonomic nomenclature follows Lesica (2002).

Analyses

We identified potential differences in the two seedling locations (gap versus matrix) using multiple response permutation procedures (MRPP) and polar ordination (Bray–Curtis) (PC-ORD v. 4.20; McCune and Grace 2002). MRPP is a nonparametric test for differences in species composition between predetermined groups of transects. Similarly, polar ordination arranges transect data, based on composition, along two predetermined endpoints (gap and matrix). It was used to potentially detect more subtle, but not necessarily significant, clusters of transect data in ordination space. Canonical correspondence analysis (CCA) was used to identify associations between species and site characteristics. In the ordination procedures, the original Bray–Curtis method was used for endpoint selection because the groups (gap and matrix) were not separated quantitatively by environmental variables. The Sørensen distance measure was used in each procedure.

We used multivariate analysis of variance (MANOVA) to identify factors associated with differences in species proportions or density counts (standardized to per hectare and then placed into proportion or density classes or bins ($n = 20$ bins)) in comparisons of gap to matrix seedlings, gap composition, substrate preference (log, soil, and stump), and height class (seedling, subcanopy, canopy, and survivor). Factors included in all MANOVA procedures were tree species, location and layer (gap seedling, matrix seedling, and matrix tree), and transect. The additional factor of seedling substrate (soil, log, and stump) was also included in analyses for seedling substrate preference, and tree height classes were included as a factor for analyses of proportions of each species in the different matrix layers. The density of seedlings was adjusted

in the analyses of substrate preference because soil and wood occurred in different proportions on the forest floor. Density counts for seedlings occurring on wood were multiplied by the ratio of soil cover to wood cover (soil/wood), while the density counts for seedlings occurring on soil were multiplied by the inverse of that ratio (wood/soil).

Several site factors that have the potential to influence establishment were included as covariates for all models. These included aspect (degrees), slope (degrees), elevation (metres), and understory percent cover categories (where seedlings were included). In the analyses for gaps, gap area (square metres) and gap aperture (degrees) were also included as covariates. Values for aperture were the angle from the center of the gap to the top of the surrounding matrix trees. Values for aspect were transformed as suggested by Stage (1976). Expanded gap area was used in our analyses instead of the canopy gap area to better reflect the portion of the forest floor actually affected by canopy openings (Gray et al. 2002). Tukey's method (95% confidence) was used to separate groups in the analyses. Significance levels were 0.05 for all analyses.

Results

Transect summary statistics

Total tree species composition consisted of 10 species: eight conifers (one deciduous) and two deciduous angiosperms (Lesica 2002) (Table 1). Summary statistics were calculated on a per hectare basis from belt transect data (matrix transects = 600 m², gap transects = 25–111 m²). An average of only five species was identified per transect (evenness = 0.630, Shannon diversity = 0.992 [McCune and Grace 2002]). Nine matrix seedling species were identified, missing only *Pinus contorta* var. *latifolia* (mean richness = 2.5 species, evenness = 0.482, Shannon diversity = 0.472 per transect), and seven gap seedling species were identified, missing *Acer glabrum* Torr. (Rocky Mountain maple), *Betula papyrifera* Marshall (paper birch), and *Larix occidentalis* Nutt. (western larch) (mean richness = 1.9, evenness = 0.485, Shannon diversity = 0.408 per gap) (Table 1). The forest matrix trees (including subcanopy trees) had a more even distribution of species as compared with seedlings in either gaps or the matrix. *Thuja plicata* and *Tsuga heterophylla* dominated each location and layer. Overall, 13.5% of the canopy was gap (including gap types not sampled).

Species composition and proportional abundance

The seedling layer

Species composition was similar between gap seedlings and matrix seedlings, and *Thuja plicata* showed no preference between the two locations. Both MRPP ($A = -0.003$, $p = 0.565$) and Bray–Curtis (polar) ordination failed to identify differences between the two groups. The first two ordination axes extracted 58.5% (axis 1 = 32.9% and axis 2 = 25.6%) of the variation in the original species matrix, but with no obvious clusters in ordination space that included predominantly matrix or gap seedlings on transects (Fig. 2). However, almost half of the total species occurred in only either the gap or the matrix seedling layer. Interestingly, *Acer glabrum*, *Betula papyrifera*, and *Larix occidentalis* seedlings, all of which are relatively shade intolerant, occurred only under the canopy of the matrix forest, and *Pinus contorta* var. *latifolia* seedlings, which are also shade intolerant, were only found in gaps (Table 1). Despite the fact that these species were indicators of a particular location (gap or matrix) within the stands, their abundance was not sufficient to allow us to reject the hypothesis of no difference in within-location heterogeneity in an MRPP setting or in ordination space.

Subcanopy and tree layers

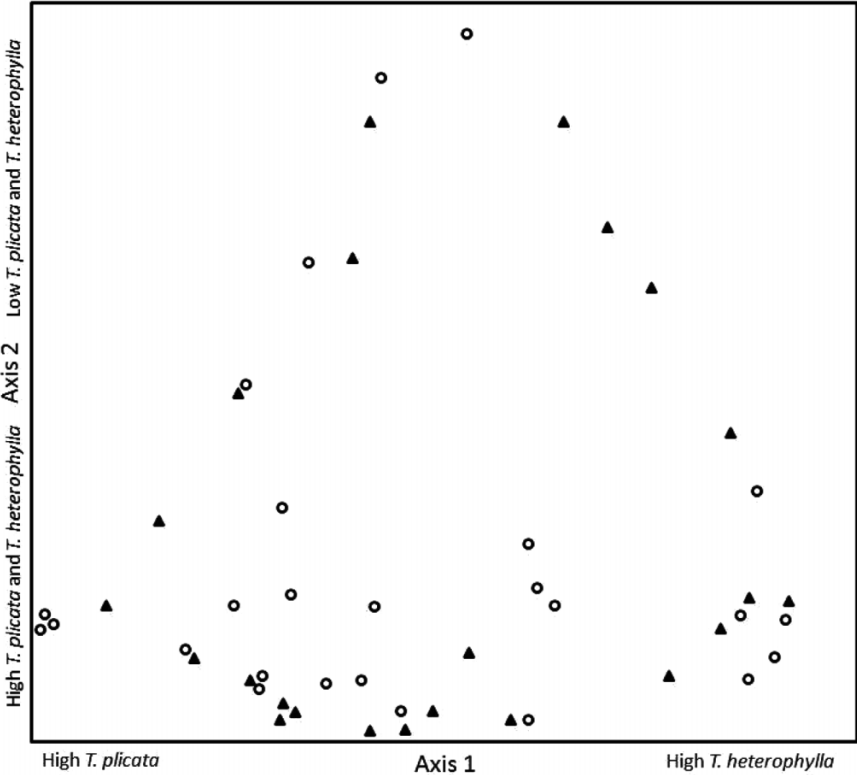
Although there were slight differences in the proportions of *Thuja plicata* among gap seedlings, matrix seedlings, and matrix

Table 1. Mean Shannon diversity, evenness, and species richness values are the averaged per transect (*n* indicated) values by matrix tree, matrix seedling, and gap seedling locations, as well as density and species proportions by these locations.

	Matrix tree (<i>n</i> = 24)		Matrix seedling (<i>n</i> = 24)		Gap seedling (<i>n</i> = 28)	
Mean diversity	0.992		0.472		0.408	
Mean evenness	0.630		0.482		0.485	
Mean richness	5.0		2.5		1.9	
Species	Matrix tree (<i>n</i> = 24)		Matrix seedling (<i>n</i> = 24)		Gap seedling (<i>n</i> = 28)	
	Density/ha	Proportion (%)	Density/ha	Proportion (%)	Density/ha	Proportion (%)
<i>Abies lasiocarpa</i>	43.7	2.1	38.9	1.4	151.2	3.2
<i>Acer glabrum</i>	0.7	0.0	2.8	0.1	0.0	0.0
<i>Betula papyrifera</i>	35.4	1.7	1.4	0.1	0.0	0.0
<i>Larix occidentalis</i>	95.7	4.6	1.4	0.1	0.0	0.0
<i>Pinus contorta</i> var. <i>latifolia</i>	120.8	5.8	0.0	0.0	9.2	0.2
<i>Picea engelmannii</i>	32.4	1.6	10.4	0.4	5.2	0.1
<i>Pinus monticola</i>	2.8	0.1	11.8	0.4	8.6	0.2
<i>Pseudotsuga menziesii</i>	110.6	5.3	14.6	0.5	54.1	1.1
<i>Thuja plicata</i>	821.0	39.6	1376.9	49.9	1659.3	34.7
<i>Tsuga heterophylla</i>	810.1	39.1	1300.7	47.1	2896.7	60.5

Note: Density values listed are the average of the individual per hectare density values for each transect, but proportion values listed are the proportion of each species by location, not by transect (sum = 100).

Fig. 2. Ordination of seedling locations. Ordination results from a Bray–Curtis (polar) ordination using species composition of seedling positions found both in the forest matrix and in gaps. Triangles represent matrix seedling and circles represent gap seedling. Neither matrix nor gap transects clustered in any portion of ordination space or along either of the axes.



trees (includes subcanopy trees), none were significant. A two-factorial MANOVA of species proportion data that included gap seedlings, matrix seedlings, and matrix trees indicated that the factors species ($p < 0.001$) and location/layer ($p = 0.001$) provided the best model (model $R^2 = 0.59$) (Table 2, model 1). The species term indicates that the proportions of individual species varied among matrix trees, matrix seedlings, and gap seedlings, while the location/layer term indicates that there was also variance in proportions among locations and layers regardless of species. The covariates (aspect, slope, elevation, and understory cover) did not

significantly affect the observed variance. *Thuja plicata* and *Tsuga heterophylla* strongly influenced these results and were dominant among all locations and layers. Groups identified using Tukey's method separated these two species from others within all locations and layers, but the locations and layers were not separated from one another between these two species. The proportion of *Thuja plicata* was highest in the gap seedlings (46.6%) and matrix seedlings (45.6%) and lowest in matrix trees (37.6%). *Tsuga heterophylla* proportions were highest in matrix trees (49.0%) followed by the matrix seedlings (47.8%) and gap seedlings (45.1%).

Table 2. MANOVA table listing factors, model R^2 values, degrees of freedom, F values, and p values for all significant MANOVA tests run.

MANOVA model	Factor					Model R^2
	Species	Location/ layer	Substrate	Successional status	Interaction	
Model 1: Species composition						
df	9	1				0.59
F	78.77	10.56				
p	<0.001	0.001				
Model 2: Substrate preference						
df	10		2		20	0.34
F	43.54		21.76		18.45	
p	<0.001		<0.001		<0.001	
Model 3: Postfire status						
df	9			3	27	0.56
F	100.78			11.29	8.09	
p	<0.001			<0.001	<0.001	

Note: These models represent the best model (highest R^2) found after testing all possible combinations of factors.

Species other than *Thuja plicata* and *Tsuga heterophylla* had higher proportional abundance values as matrix trees rather than matrix seedlings or gap seedlings, except *Pinus monticola*, which had its highest proportional abundance in matrix seedlings, and *Abies lasiocarpa* (Hook.) Nutt. (subalpine fir), which was highest in gap seedlings.

Gap seedling composition and site characteristics

A CCA of the gap seedling composition matrix and the gap site characteristic matrix indicated that 45.5% of the variance in composition within gaps alone was explained by two axes (Fig. 3). Axis 1 (29.2% of variance explained) was most associated with gap shrub cover, elevation, and gap area. Axis 2 (16.2% of variance explained) was associated with gap aperture, although the vector for this variable was not entirely parallel with the axis. Even though transects were fairly tightly clustered in one area of ordination space, gradients were still observed. *Pinus contorta* var. *latifolia*, *Pseudotsuga menziesii* (Mirb.) Franco (Douglas-fir), and *Abies lasiocarpa* were plotted in the area of the ordination associated with smaller gap apertures (lower surrounding canopy height) and smaller gap areas, while *Thuja plicata*, *Tsuga heterophylla*, and *Pinus monticola* were in ordination space associated with larger gap area and larger apertures. *Abies lasiocarpa* and *Picea engelmannii* Parry ex Engelm. (Engelmann spruce) were more common in high-elevation gaps.

Seedling substrate preference

In the MANOVA for seedling substrate preference, location (gap or matrix) within the stand did not contribute significantly to the model ($p \leq 0.184$), indicating that substrate preferences of species do not differ between the matrix and gaps. The best model included species, substrate, and the interaction term species $\times$ substrate ($p < 0.001$ for each, model $R^2 = 0.34$) (Table 2, model 2). While there were no differences between the matrix seedlings and gap seedlings, all substrate groups were separated (Tukey's method). Logs and stumps contained the highest proportion of seedlings (59%) and soil contained about 41% of seedlings (Table 3). *Thuja plicata* and *Tsuga heterophylla* were the most abundant species (95.5% of total) and were primarily found on wood, with the other species barely represented in the seedling layer (4.5% of total). Differences among all other species–substrate combinations were insignificant, but other species were more commonly found on soil than on wood. Adding covariates did not increase the R^2 values (all significant R^2 increases < 0.01). Of the seedlings that established on wood, all established on well-decayed wood (class 4 or 5 (Narukawa et al. 2003)).

Postfire species status

To identify differences in successional status among species, MANOVA was run using height classes, species, and the interaction term height class $\times$ species as factors in the best model (all $p < 0.001$, $R^2 = 0.56$) (Table 2, model 3). No covariates affected the variance explained by the model (all $p \geq 0.273$). The height class factor alone indicated that fire survivor was the only class to differ significantly in species proportions because *Larix occidentalis* dominated the fire survivor class (mean proportion = 19.1%), while all others were dominated by *Thuja plicata* and *Tsuga heterophylla*. Groups formed by the Tukey test on species separated *Thuja plicata* and *Tsuga heterophylla* as a group (highest proportions), *Larix occidentalis*, *Pseudotsuga menziesii*, *Picea engelmannii*, and *Abies lasiocarpa* as a group, and *Pseudotsuga menziesii*, *Picea engelmannii*, *Abies lasiocarpa*, *Betula papyrifera*, *Pinus contorta* var. *latifolia*, and *Pinus monticola* as a group, and *Acer glabrum* was alone (lowest proportion). The interaction term further elucidated the relationship among tree height classes for each species. Postfire canopy tree, subcanopy, and seedling proportions were equal among these three classes for both *Tsuga heterophylla* (mean proportion for postfire canopy tree = 41.8%, subcanopy = 53.0%, seedling = 48.1%) and *Thuja plicata* (mean proportion for postfire canopy tree = 32.9%, subcanopy = 37.9%, seedling = 44.7%). *Tsuga heterophylla* dominated all three classes followed by *Thuja plicata*. All other proportions were less than 20%.

Discussion

In GNP, *Thuja plicata* established just as readily following fire as beneath the canopy and in gaps in mature forests. Despite low levels of regeneration in locations near GNP, which might suggest a retreat of the eastern range margin (Habeck and Mutch 1973), our study demonstrated that *Thuja plicata* is present in all successional stages at the easternmost extent of its distribution. Gap and matrix seedlings were dominated by *Thuja plicata* and *Tsuga heterophylla* followed by *Abies lasiocarpa*, all species common to later-seral stages. Likely because of the relatively young age of these stands, total canopy gap area was low (13%) as compared with older stands. For example, in *Pseudotsuga menziesii* – *Tsuga heterophylla* stands in Washington, the gap area was over twice as large (33.6%) as was found in our sites (Spies et al. 1990). As the proportion of the canopy in gaps increases, less shade-tolerant species may occasionally be permitted to establish (Kneeshaw and Bergeron 1998).

Factors that contribute to differences in distribution and establishment success such as aspect and slope were not identified as significant in our ANOVA tests. This likely reflects the relatively small scale at which we conducted our sampling. At larger scales

Fig. 3. Ordination of gap characteristics. Canonical correspondence analysis of gap seedling composition (main matrix) and gap characteristics (second matrix). Axis 1 was most associated with gap shrub cover, elevation, and gap area, while axis 2 was associated with gap aperture. Plus signs represent species location in ordination space and triangles represent transects.

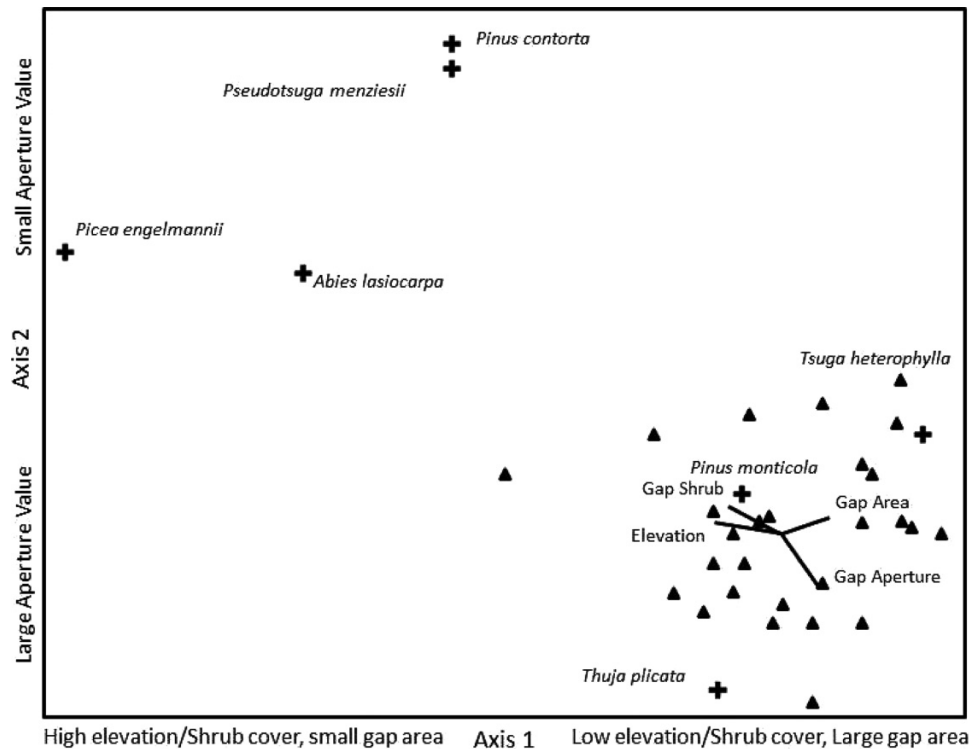


Table 3. Average number (per hectare) of seedlings in each of the seedling locations (matrix and gap) per transect (*n* indicated) as well as the proportion of establishment substrates by species in each location.

Species	Matrix (<i>n</i> = 24)						Gap (<i>n</i> = 28)					
	Average (density/ha)			Proportion (%)			Average (density/ha)			Proportion (%)		
	Log	Stump	Soil	Log	Stump	Soil	Log	Stump	Soil	Log	Stump	Soil
<i>Abies lasiocarpa</i>	0	0	7.9	0	0	<0.01	0	0	61.7	0	0	0.03
<i>Acer glabrum</i>	0	0	0.30	0	0	<0.01	0	0	0	0	0	0
<i>Betula papyrifera</i>	0	0	0.16	0	0	<0.01	0	0	0	0	0	0
<i>Larix occidentalis</i>	0	0	0.19	0	0	<0.01	0	0	0	0	0	0
<i>Pinus contorta</i> var. <i>latifolia</i>	0	0	0	0	0	0	0	0	5.2	0	0	<0.01
<i>Picea engelmannii</i>	10.5	0	1.5	<0.01	0	<0.01	0	0	2.2	0	0	<0.01
<i>Pinus monticola</i>	72.6	0	0.78	0.01	0	<0.01	0	0	1.6	0	0	<0.01
<i>Populus tremuloides</i>	0	0	0	0	0	0	0	0	0	0	0	0
<i>Pseudotsuga menziesii</i>	0	0	2.5	0	0	<0.01	0	0	16.9	0	0	0.02
<i>Thuja plicata</i>	610.9	128.4	186.0	0.18	0.02	0.19	3539.4	1127.9	552.2	0.11	0.05	0.27
<i>Tsuga heterophylla</i>	3540.7	167.8	128.9	0.53	0.03	0.04	14552.8	1110.8	1066.7	0.34	0.03	0.15

Note: These values are adjusted for the differences in area covered by each substrate type. Unadjusted values would indicate that more seedlings were found growing on soil than on logs or stumps.

that extend beyond the range margin of these species, each of these would likely have been a good predictor of differences among species. The ordination tests begin to hint at this, at least with the elevation variable.

Gap and matrix seedling establishment

In our stands, gap-filler composition was strongly affected by canopy species composition (spatially contingent succession) to the exclusion of species that did not dominate the canopy (Baker 2009). In GNP, only *Pseudotsuga menziesii* and *Tsuga heterophylla* have similar longevity in this forest zone, and given enough time, *Thuja plicata* and *Tsuga heterophylla* could use canopy gaps to attain canopy and stand dominance throughout stand maturation because they were present in relatively high proportions within gaps and below the canopy and are shade tolerant, while *Pseudotsuga menziesii* is not.

Spies et al. (1990) found sapling density in mature *Pseudotsuga menziesii* forests in the Cascade Range to be lower than we found in GNP for *Tsuga heterophylla* and *Thuja plicata*. Although we included seedlings in our counts of gap and understory establishment, we found a much higher rate of establishment under the canopy for *Thuja plicata* and a comparable rate for *Tsuga heterophylla*. In gaps, the number of *Tsuga heterophylla* observed in GNP was almost double that observed in the Cascade Range (Spies et al. 1990), and *Thuja plicata* numbers were almost an order of magnitude higher in GNP. Both of these species are considered mesophytic, and the GNP region is drier than the Cascade Mountain Range. Even so, regeneration appears abundant.

In a southern range-margin location in the Selway–Bitterroot Wilderness in northern Idaho, Habeck (1978) found very little to

no *Thuja plicata* seedling establishment at elevations below 1100 m but significant establishment (3501 seedlings/ha) above this elevation in stands over 315 years old. Our transects were only slightly higher in elevation (up to 1356 m elevation), were younger, and had fewer seedlings per hectare in both gaps and the matrix. However, as stands open through senescence of canopy individuals, conditions will become more gap-like within more of the matrix, and it is within gaps that we saw increased numbers of *Thuja plicata* seedlings in our stands. GNP represents the eastern range margin for *Thuja plicata*, and the population is presumably limited by low temperatures at higher elevations as it approaches the Continental Divide. Elevation was important in the ordination of community composition, and *Thuja plicata* was plotted in the lower elevation side of ordination space (Fig. 3). Had we placed more emphasis on riparian stands of *Thuja plicata* at the elevational extremes of the range margin, we might have observed patterns similar to those seen in the southern limits of the species. In the Selway–Bitterroot, *Thuja plicata* primarily inhabits riparian corridors, and *Pseudotsuga menziesii*, *Abies lasiocarpa*, and *Picea engelmannii*, instead of *Thuja plicata*, dominate higher elevations (Habeck and Mutch 1973).

Thuja plicata is shade tolerant, but *Tsuga heterophylla* is likely the most shade tolerant of all tree species encountered in GNP, and this is likely reflected in its success and relative dominance at multiple successional stages. *Pseudotsuga menziesii*, a shade-intolerant species (Maas-Hebner et al. 2005), showed relatively low rates of establishment in the matrix or in gaps, where it was not part of the initial postfire cohort in our stands. In GNP, gap size, which relates to available light, was not a statistically significant factor in determining the suitability of a gap establishment site for *Thuja plicata* as has been found in other studies (Spies et al. 1990), possibly because seedlings were present as advance regeneration before the gap was formed. *Thuja plicata* was, however, associated with larger gaps as indicated by CCA, suggesting that gap size does play at least a small role in regeneration success. A longer fire interval, which would allow for the creation of larger tree-fall gaps as forests mature and begin to experience tree mortality, may allow species such as *Pseudotsuga menziesii* and *Larix occidentalis* to establish within gaps (Coates 2002). However, when early-seral stages are dominated by shade-tolerant species, intolerant species such as *Betula papyrifera* are disadvantaged throughout stand development, as shown in our study.

The ability of *Thuja plicata* in GNP to establish as a relatively large component of the postfire cohort and its ability to use gaps and subcanopy locations for regeneration affect its survival on the landscape, likely through an adaptation to range-margin conditions. These conditions vary locally and differ from other areas where *Thuja plicata* is more abundant but behaves more as a late-successional species (Ishii et al. 2000; Larson and Franklin 2005). Because species in different forest locations were not separated in ordination space, communities are considered similar between gaps and the canopy matrix, results observed in other gap–matrix community comparisons (de Römer et al. 2007). However, *Thuja plicata* seedlings have also been observed to establish in very low numbers beneath the canopy of mixed-species forests in British Columbia, which is a more central location within the range of the species (Coates 2002).

Substrate preference

Forest tree species and understory species preferentially utilize different substrates for establishment because substrates differ in nutrient and moisture levels (Yavitt and Fahey 1985; Kennedy and Quinn 2001). As in more western populations where *Thuja plicata* and *Tsuga heterophylla* are more frequently encountered, both species were found having established on fallen and decaying logs and stumps (Christy and Mack 1984; Stewart 1989). These findings represent density values that were spatially corrected and indicate a preference for one substrate over another. However, seed-

ling density without spatial correction was higher on soil in our study, simply as a function of the abundance of soil versus decayed wood. In the PNW, *Tsuga heterophylla* established more frequently on fallen logs (Franklin and Hemstrom 1981), as it did in GNP. A similar effect was seen with *Thuja plicata* on the Olympic Peninsula, Washington, where, although differences among substrate preferences were not significant, more *Thuja plicata* seedlings established on logs than on soil (Harmon and Franklin 1989). Harmon and Franklin (1989) thought persistent standing water might limit seedlings to logs, but water was not a significant factor in their study. Our transects were not near standing water, and raised wood will dry more quickly than soil, but the range in GNP where *Thuja–Tsuga* forest occurs is drier but still climatically similar to the PNW, and wood must remain sufficiently moist for seedling growth.

The only other *Thuja* species in North America (*Thuja occidentalis* L. (northern white cedar)) is documented to prefer moss-covered logs over logs without moss or soil for establishment in Quebec, Canada (Simard et al. 1998). However, *Thuja plicata* seedlings on Vancouver Island (British Columbia) survived best on mineral soil as opposed to soil covered with litter or in sites with partial vegetation removal (Weber et al. 2003). Although we did not separate our substrates into categories based on cover (moss), our observations in the field suggest that *Thuja plicata* in GNP established less frequently on moss-covered logs and when establishing in soil to have done so on litter, not on exposed mineral soil. All other species encountered were found more often on wood, but their occurrence was too infrequent to affect overall results from the model.

Postfire *Thuja plicata* regeneration

Thuja plicata is typically thought to be a late-successional, shade-tolerant species (Carter and Klinka 1992; Kobe and Coates 1997), establishing in a stand through gap-phase regeneration and less so immediately following fire (Wright et al. 1998; Larson and Franklin 2005). Further, it is easily killed by fire through stem girdling (Peterson and Ryan 1986). In clearcut stands, Lutz and Halpern (2006) found *Thuja plicata* establishment to occur sooner than expected (<40 years) but to still be relatively low as compared with other co-occurring species, patterns seen in a location close to GNP (Swan Valley, Montana) as well (Antos and Habeck 1981). However, in GNP, *Thuja plicata* has been observed behaving more as a pioneer species, colonizing disturbed areas immediately after the disturbance (Shearer and Asebrook 2003). Tree-ring widths from trees used in our study were wide in the oldest rings (D.A. McKenzie and D.B. Tinker, unpublished data), narrowing as one moves farther from the pith. This pattern is indicative of open, unrestricted growth early in the life of the tree, conditions found following fire and not the light-limited conditions of the subcanopy (Daniels 2003). Our study indicated that *Thuja plicata* successfully regenerated both in the understory of the closed canopy and within gaps. The postfire conditions created by stand-replacing fires create conditions very different from those found in later-successional stages to which survivors and new recruits alike must be adapted (Swanson et al. 2011). Based on our results, *Thuja plicata* is suited not only to subcanopy and gap regeneration but also to postfire conditions in many areas in the park, allowing it to establish after fire in higher proportions than found elsewhere (Hamilton 2006).

Although the current study was designed for smaller canopy gaps and mature forests, we also had the opportunity to characterize immediate postfire regeneration in the *Thuja–Tsuga* zone of GNP (D.A. McKenzie and D.B. Tinker, unpublished data). The stand initiated after the 2003 Robert Fire on the northwest side of Lake McDonald was 7 years old at the time of the current study. We placed transects at the north end of Lake McDonald in areas of moderate to high burn severity, but low-severity burned areas were within 500 m of all three transects. Seedling abundance was

patchy, but mean seedling density was still 207 511/ha, much higher than that found within the other study sites. *Thuja plicata* represented 22.1% and *Tsuga heterophylla* represented 73.7% of the seedling layer. *Picea engelmannii* (3.5%) was the only other species with a proportion greater than 1%. For *Thuja plicata* and *Picea engelmannii*, over 96% of seedlings were found growing on soil, and for *Tsuga heterophylla*, 99.9% of seedlings were on soil. The proportion of *Thuja plicata* and *Tsuga heterophylla* is not surprising given that the majority of the surviving trees in the nearby lightly burned area were of these two species. The high proportion of seedlings growing in soil as opposed to logs may reflect the drier postfire conditions present here, as opposed to within a closed canopy or small gap.

We also sampled two sites within the same fire perimeter at the south end of Lake McDonald in a previous study (McKenzie and Tinker 2012). These were 10 km distant from the above-mentioned sites, but postfire regeneration of *Thuja plicata* was very different. These southern transects experienced high-intensity fire for at least 1 km in all directions. Here, prefire relative density of *Thuja plicata* was 22.9%, but postfire density was less than 1%. The prefire relative density of *Thuja plicata* was somewhat lower than that observed in the 2010 tree data, but the lack of postfire regeneration by *Thuja plicata* was not even suggested by the current data. *Thuja plicata* does not have a long-lived seed bank, seeds are not fire tolerant, nor are they widely dispersed (Minore 1983), and the differences between the two sites point to fire-surviving trees as an essential part of *Thuja plicata* regeneration success following disturbance by fire.

Seedlings of *Thuja plicata* and *Tsuga heterophylla* were most dense in moist locations, suggesting that when they establish in full sunlight, abundant moisture is required for successful development (Weber et al. 2003). Younger trees are considered to be moderately exposure tolerant, but seedlings are susceptible to sun scorching, establishing better in shade (Wright et al. 1998). However, observation of the north site suggests that this is not necessarily true in GNP where there was a seed source as well as abundant direct solar radiation. This has been further documented along road construction sites within GNP where, despite overall low germination rates, *Thuja plicata* and *Tsuga heterophylla* made up 95% of the postconstruction, naturally regenerating seedling community (Shearer and Asebrook 2003). The ability of *Thuja plicata* to occupy all successional stages and tolerate the wide range of environmental conditions therein suggests that at its range margin, *Thuja plicata* is very much a generalist and is able to acclimate or has adapted to a wide range of conditions.

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